

HarvardX Capstone Project - Recommendation System

Frank Valdivia

2025-08-05

Contents

1	INTRODUCTION	2
2	ANALYSIS AND METHODS	2
2.1	loading libraries	2
2.2	Loading files from server and generating datasets	2
2.3	Analysis and methods	5
2.3.1	Root Mean Squared Error (RMSE)	5
2.3.2	First model: $Y = \mu_hat$	6
2.3.3	Second model: $Y = \mu_hat + Movie_bias$	7
2.3.4	Third model: $Y = \mu_hat + Movie_bias + User_bias$	9
2.3.5	Fourth model: $Y = \mu_hat + Movie_Regularized_bias$ (Regularization)	11
2.3.6	Fifth model: $Y = \mu_hat + Movie_Regularized_bias$ (Regularization) with Optimized Lamda	15
2.3.7	Sixth model: $Y = \mu_hat + Movie_Regularized_bias + User_Regularized_bias$ (Regularization) with Optimized Lamda	16
3	RESULTS	18
4	CONCLUSION	19

1 INTRODUCTION

The goal of this project is to build a recommendation system for Netflix. This recommendation system would predict which movies users would like to watch. Based on the history of movie ratings by users, the system can suggest movies the users might be most interested in.

In this project, the MovieLens dataset from Netflix will be used to create the recommendation system. A subset of 10 million ratings will be used to reduce computing time.

Machine learning methods will be used, which means that two sub datasets will be created during this process:

- A train set will be generated with the name: `edx` - this dataset will be used to train the model.
- A test set will be generated with the name: `final_holdout_test` - this dataset will be used to test the model trained using the train dataset.

The Root Mean Squared Error (RMSE) will be used to calculate the error between predictions and true values in the train set (`edx` set)

During this process, several sub datasets will be created and are explained in the Methods section.

2 ANALYSIS AND METHODS

The overall process is as follows:

- 2.1. Loading libraries
- 2.2. Loading files from server and generating datasets
- 2.3. Analysis and methods

2.1 loading libraries

To run the report the following R libraries need to be previously installed:

- `library(tidyverse)`
- `library(caret)`
- `library(ggplot2)`
- `library(dplyr)`
- `library(gridExtra)`
- `library(beepr)`
- `library(stringr)`
- `library(readr)`

In this step above libraries are loaded

2.2 Loading files from server and generating datasets

The files are downloaded, unzipped and used to create data sets.

Movies Ratings and Movies are pulled from:

<http://files.grouplens.org/datasets/movielens/ml-10m.zip>

The following datasets are created and will be used in training the different models.

- `movielens`: Movie ratings including Movie information

- ratings: Movie ratings with MovieId and UserId
- movies: Movie information
- edx: Train dataset to be used to train every model
- final_holdout_test: Test dataset to be used to test the selected model

Following are the structure and number of rows that each of these datasets have:

##

Movielens Dataset: Head

userId	movieId	rating	timestamp	title	genres
1	122	5	838985046	Boomerang (1992)	Comedy Romance
1	185	5	838983525	Net, The (1995)	Action Crime Thriller
1	231	5	838983392	Dumb & Dumber (1994)	Comedy
1	292	5	838983421	Outbreak (1995)	Action Drama Sci-Fi Thriller
1	316	5	838983392	Stargate (1994)	Action Adventure Sci-Fi
1	329	5	838983392	Star Trek: Generations (1994)	Action Adventure Drama Sci-Fi

##

Movielens Dataset: Number of Records

$$\frac{\dim(\text{movielens})}{10000054}$$

##

Ratings Dataset: Head

userId	movieId	rating	timestamp
1	122	5	838985046
1	185	5	838983525
1	231	5	838983392
1	292	5	838983421
1	316	5	838983392
1	329	5	838983392

##

Ratings Dataset: Number of Records

$$\frac{\dim(\text{ratings})}{10000054}$$

##

Movies Dataset: Head

movieId	title	genres
1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
2	Jumanji (1995)	Adventure Children Fantasy
3	Grumpier Old Men (1995)	Comedy Romance

movieId	title	genres
4	Waiting to Exhale (1995)	Comedy Drama Romance
5	Father of the Bride Part II (1995)	Comedy
6	Heat (1995)	Action Crime Thriller

##

Movies Dataset: Number of Records

dim(movies)
10681

##

edx (train set) Dataset: Head

	userId	movieId	rating	timestamp	title	genres
1	1	122	5	838985046	Boomerang (1992)	Comedy Romance
2	1	185	5	838983525	Net, The (1995)	Action Crime Thriller
4	1	292	5	838983421	Outbreak (1995)	Action Drama Sci-Fi Thriller
5	1	316	5	838983392	Stargate (1994)	Action Adventure Sci-Fi
6	1	329	5	838983392	Star Trek: Generations (1994)	Action Adventure Drama Sci-Fi
7	1	355	5	838984474	Flintstones, The (1994)	Children Comedy Fantasy

##

edx (train set) Dataset: Number of Records

dim(edx)
9000055

##

final_holdout_test (test set) Dataset: Head

	userId	movieId	rating	timestamp	title	genres
1	231	5	838983339	2	Dumb & Dumber (1994)	Comedy
1	480	5	838983653	2	Jurassic Park (1993)	Action Adventure Sci-Fi Thriller
1	586	5	838984068	2	Home Alone (1990)	Children Comedy
2	151	3	868246450	2	Rob Roy (1995)	Action Drama Romance War
2	858	2	868245646	2	Godfather, The (1972)	Crime Drama
2	1544	3	868245920	2	Lost World: Jurassic Park, The (Jurassic Park 2) (1997)	Action Adventure Horror Sci-Fi Thriller

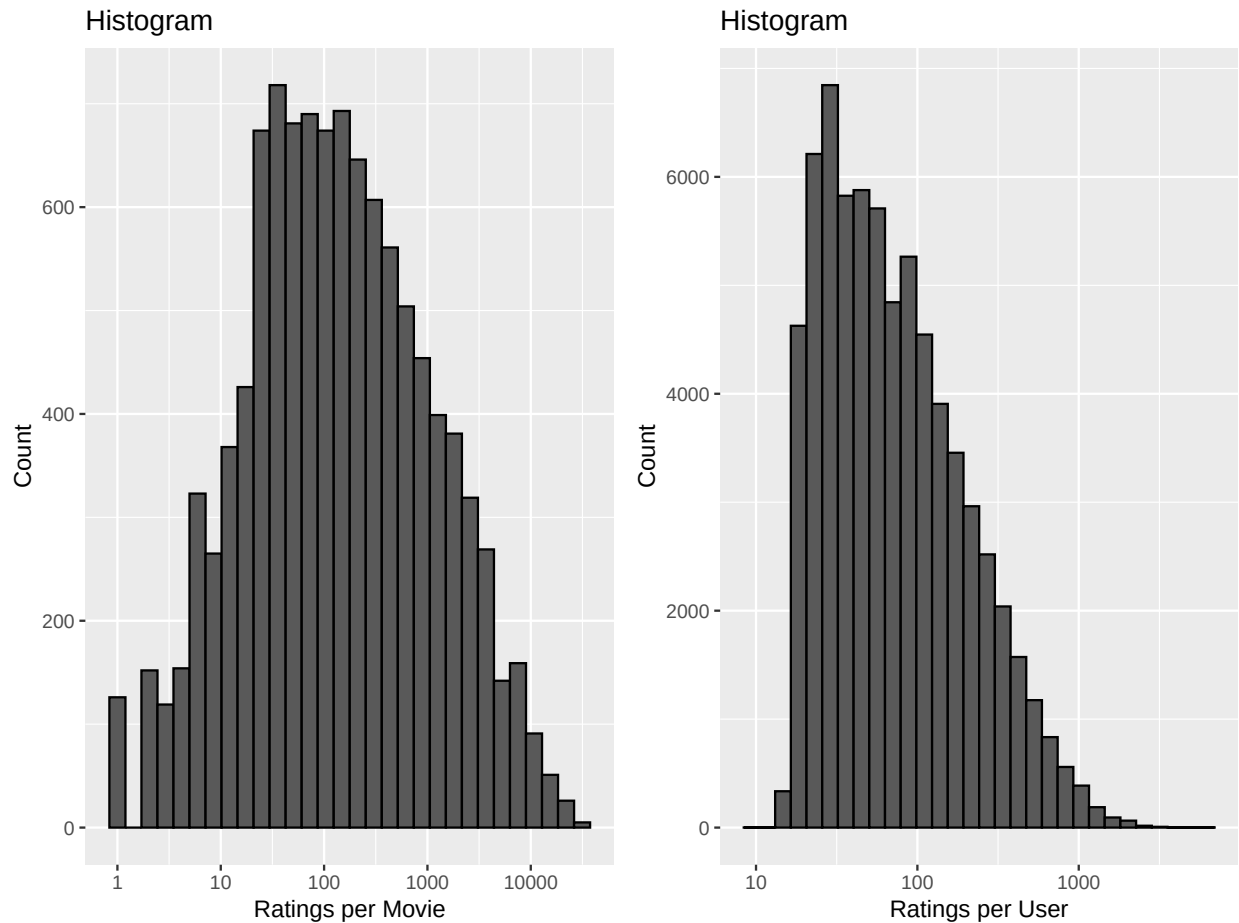
##

final_holdout_test (test set) Dataset: Number of Records

dim(final_holdout_test)
999999

2.3 Analysis and methods

Following are the Histograms of Ratings counted by Movie and by User taken from the Train dataset (edx)



Some general characteristics of the data:

- They seem to be normally distributed.
- Some movies have more ratings than others.
- Some users rate more movies than others.
- There is no significant presence of outliers.

The models will use ratings from movies, users, and movies & users.

2.3.1 Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE), which is the square root of the mean squared error (MSE), will be used to estimate the error of every method. Reference: <https://rafalab.dfci.harvard.edu/dsbook/introduction-to-machine-learning.html#loss-function>

Six models will be built and each model assessed against its value of RMSE.

2.3.2 First model: $Y = \mu_{\text{hat}}$

The predictor of the first model is just the mean (μ_{hat}) of ratings in the train set (edx)

$y = \mu_{\text{hat}}$

RMSE can be calculated using the train set (edx) and μ_{hat} as predictor.

```
## mu_hat: 3.512465
```

```
## mu_hat_rmse: 1.060331
```

The expectation is that any number other than μ_{hat} would result in a higher RMSE. A prediction and RMSE can be calculated using $\mu_{\text{hat}} + 0.1$, which is equal to $\mu_{\text{hat}} + 0.1$

```
## mu_hat+0.1: 3.612465
```

```
## mu_hat+0.1_rmse: 1.065036
```

As expected, $\mu_{\text{hat}} + 0.1$ _rmse is higher than μ_{hat} _rmse

The first model and its RMSE are as follows:

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.060331	

2.3.3 Second model: $Y = \mu_hat + \text{Movie_bias}$

Every Movie has its own Rating mean, which might be different from the overall mean or different from other Movie Rating mean. That is due to the bias (or effect) of every movie on the ratings.

In the second model, the Movie bias or effect will be incorporated.

The Bias will be calculated for every Movie as the difference between the Mean Rating of that movie and the Mean Rating of the whole dataset (μ_hat).

b_i (or b_i) will be the Bias per Movie.

b_i is then added to μ_hat for every Movie and that is the Predictor for that Movie.

$$Y = \mu_hat + b_i$$

Adding the Movie bias or b_i should improve the prediction and generate a smaller RMSE

b_i is calculated for every movie and stored into the `movie_avgs` dataset.

##

movie_avgs Dataset: Head

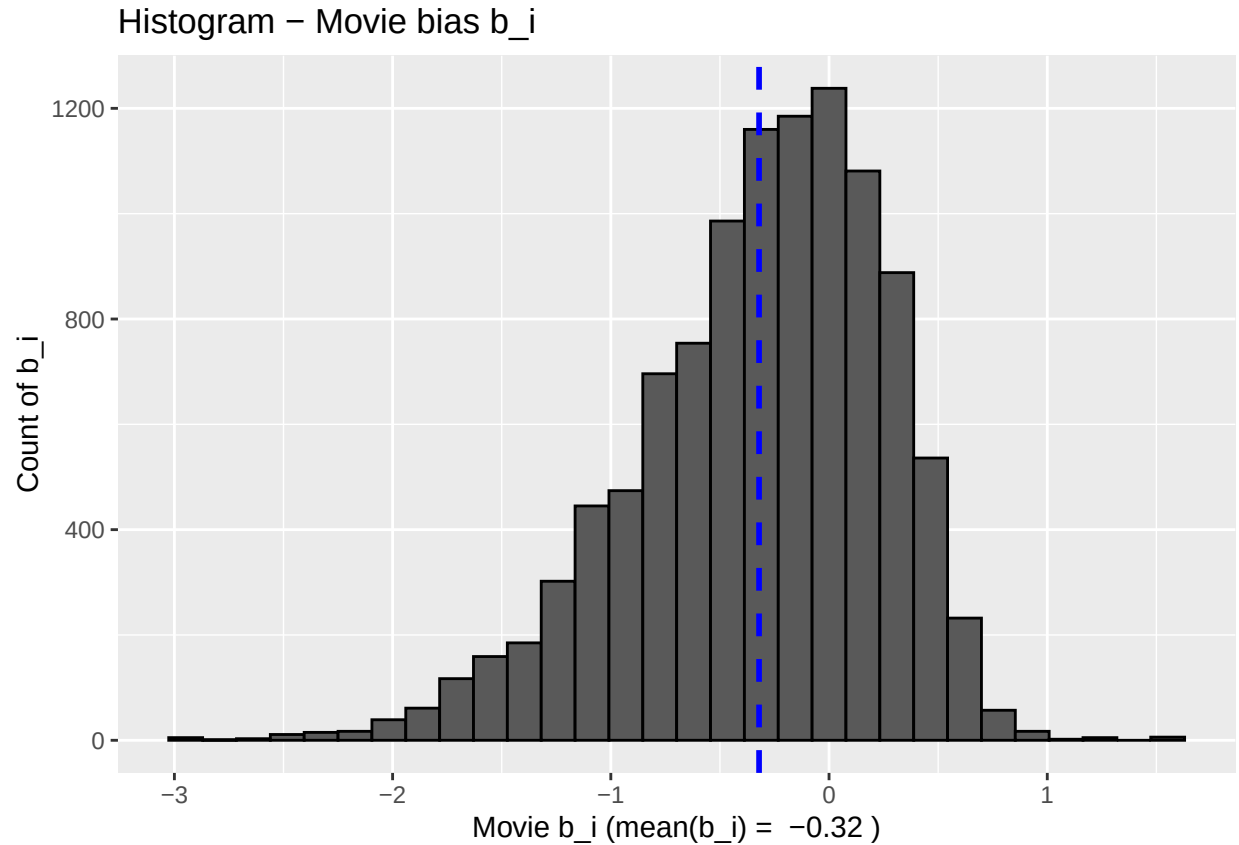
movieId	b_i
1	0.4151725
2	-0.3070658
3	-0.3654817
4	-0.6481659
5	-0.4437933
6	0.3028191

##

movie_avgs Dataset: Number of Records

dim(movie_avgs)
10677

In the following plot, the distribution of Movie bias shows that there is variation around the value of 0 and most bias numbers go from -3 to 1



When adding Movie bias to the previous model, the prediction is $Y = \mu_{\text{hat}} + \text{Movie_bi}$ where $\text{Movie_bi} = \text{Movie bias}$

RMSE can be calculated using the train set (edx) and $\mu_{\text{hat}} + \text{Movie_bi}$ as predictor.

```
## movie_bi_rmse: 0.9423475
```

The second model and its RMSE are as follows:

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.0603313	
2: Predictor = $\mu_{\text{hat}} + \text{Movie_bi}$	0.9423475	

When adding Movie bias, the RMSE is lower than the first model's RMSE

2.3.4 Third model: $Y = \mu_{\text{hat}} + \text{Movie_bias} + \text{User_bias}$

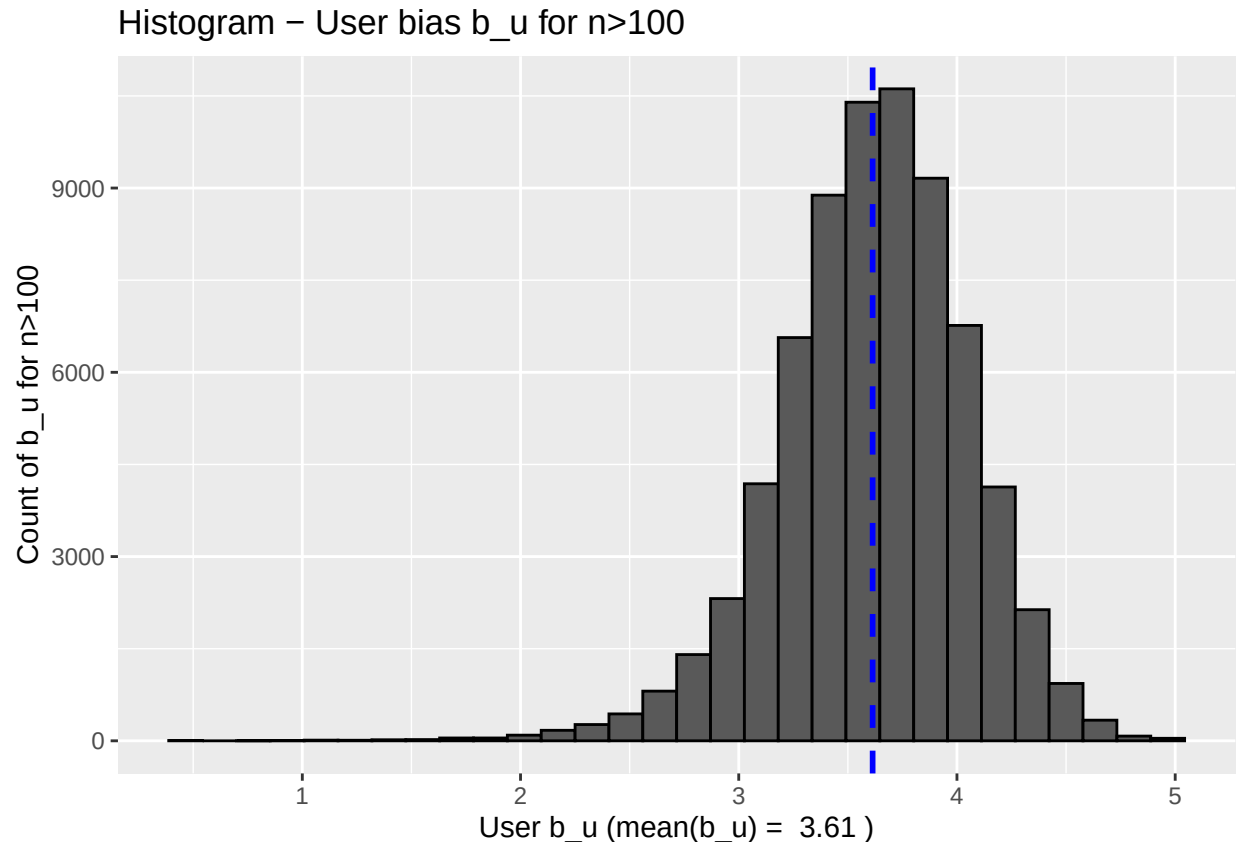
In the third model, User bias (effect) will be added to the previous model, Movie bias.

This updated model will have both Movie and User biases.

bu (or b_u) will be the Bias per User. bu is added to μ_{hat} and movie_bi for every Movie and User and that is the Predictor.

Adding the User bias to Movie bias should improve the prediction and generate a smaller RMSE

In the following plot, only Users that have rated 100 or more movies are included.



The distribution of Ratings shows that there is variation around the value of 4 and most ratings go from 1 to 5 This variation means that some users rate higher than other users do.

This variability also suggests that incorporating User bias would indeed improve the model

The third model will be $Y = U + b_i + b_u$

Here the prediction is $Y = U + \text{Movie_bi} + \text{User_bu}$

User bias will be calculated by subtracting μ_{hat} and movie_bias from Y (Rating)

$\text{User_bias} = b_u = Y - u - b_i$

User_bias will be averaged per user and stored in a new dataset: user_avgs

##

user_avgs Dataset: Head

userId	b_u
1	1.6792347
2	-0.2364086
3	0.2643303
4	0.6520781
5	0.0852677
6	0.3462454

##

user_avgs Dataset: Number of Records

dim(user_avgs)
69878

The prediction is $Y = \mu_{\text{hat}} + \text{Movie_bi} + \text{User_bu}$ where Movie_bi = Movie bias; User_bu = User bias.

RMSE can be calculated using the train set (edx) and $\mu_{\text{hat}} + \text{Movie_bi} + \text{User_bu}$ as predictor.

movie_bi_user_bu_rmse: 0.8567039

The third model and its RMSE are as follows:

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.0603313	
2: Predictor = $\mu_{\text{hat}} + \text{Movie_bi}$	0.9423475	
3: Predictor = $\mu_{\text{hat}} + \text{Movie_bi} + \text{User_bu}$	0.8567039	

When adding User bias to Movie bias, the RMSE is lower than the previous model's RMSE

2.3.5 Fourth model: $Y = \mu_{\text{hat}} + \text{Movie_Regularized_bias}$ (Regularization)

The fourth model will apply Regularization.

RMSE was reduced when movie bias and user bias were incorporated but does not take into account when there are very few ratings (small sample size) for a movie or user.

The following table shows the 15 largest mistakes predicted by the second model ($Y = \mu_{\text{hat}} + \text{Movie_bias}$). Residuals are calculated subtracting $(\mu_{\text{hat}} + \text{bi})$ from Y from the test set:

Ratings with highest absolute residuals	UserId	Residual
Pokémon Heroes (2003)	56965	3.970803
Shawshank Redemption, The (1994)	14871	-3.955131
Shawshank Redemption, The (1994)	39827	-3.955131
Shawshank Redemption, The (1994)	45000	-3.955131
Godfather, The (1972)	37651	-3.915366
Godfather, The (1972)	51578	-3.915366
Godfather, The (1972)	59583	-3.915366
Usual Suspects, The (1995)	29999	-3.865854
Usual Suspects, The (1995)	48793	-3.865854
Usual Suspects, The (1995)	53358	-3.865854
Schindler's List (1993)	30564	-3.863493
Schindler's List (1993)	48146	-3.863493
Schindler's List (1993)	51751	-3.863493
Pokemon 4 Ever (a.k.a. Pokémon 4: The Movie) (2002)	56965	3.821782
Casablanca (1942)	14159	-3.820424

##

Following is the list of the 15 best movies based just on Movie_bias.

Movie title (Best)
Hellhounds on My Trail (1999)
Satan's Tango (Sátántangó) (1994)
Shadows of Forgotten Ancestors (1964)
Fighting Elegy (Kenka erejii) (1966)
Sun Alley (Sonnenallee) (1999)
Blue Light, The (Das Blaue Licht) (1932)
Who's Singin' Over There? (a.k.a. Who Sings Over There) (Ko to tamo peva) (1980)
Human Condition II, The (Ningen no joken II) (1959)
Human Condition III, The (Ningen no joken III) (1961)
Constantine's Sword (2007)
More (1998)
I'm Starting From Three (Ricomincio da Tre) (1981)
Class, The (Entre les Murs) (2008)
Mickey (2003)
Demon Lover Diary (1980)

##

Most of the movies above are not well known.

##

Following is the number of ratings of the 15 best movies based just on Movie_bias.

[1] 1 2 1 1 1 1 4 4 4 2 7 3 3 1 1

```
##
## These unknown movies have just one or very few ratings (very small sample size),
## which does not provide a good predictor. Regularization corrects that.
##
## Following is the list of the 15 worst movies based just on Movie_bias.
```

Movie title (Worst)
Besotted (2001)
Hi-Line, The (1999)
Accused (Anklaget) (2005)
Confessions of a Superhero (2007)
War of the Worlds 2: The Next Wave (2008)
SuperBabies: Baby Geniuses 2 (2004)
Hip Hop Witch, Da (2000)
Disaster Movie (2008)
From Justin to Kelly (2003)
Criminals (1996)
Mountain Eagle, The (1926)
Stacy's Knights (1982)
Dog Run (1996)
Monkey's Tale, A (Les Château des singes) (1999)
When Time Ran Out... (a.k.a. The Day the World Ended) (1980)

```
##
## Most of the movies above are not well known.
##
## Following is the number of ratings of the 15 worst movies based just on Movie_bias.
## [1]  2  1  1  1  2 56 14 32 199  2  2  1  1  1  1
```

```
##
## These unknown movies have just one or very few ratings (very small sample size),
## which does not provide a good predictor. Regularization corrects that.
```

Good sample size of Ratings per Movie is desired to have a good predictor. Very few ratings means that those movies have very small sample sizes.

Having less ratings (smaller sample sizes) increases bias and generates higher residuals. Higher residuals increase the RMSE.

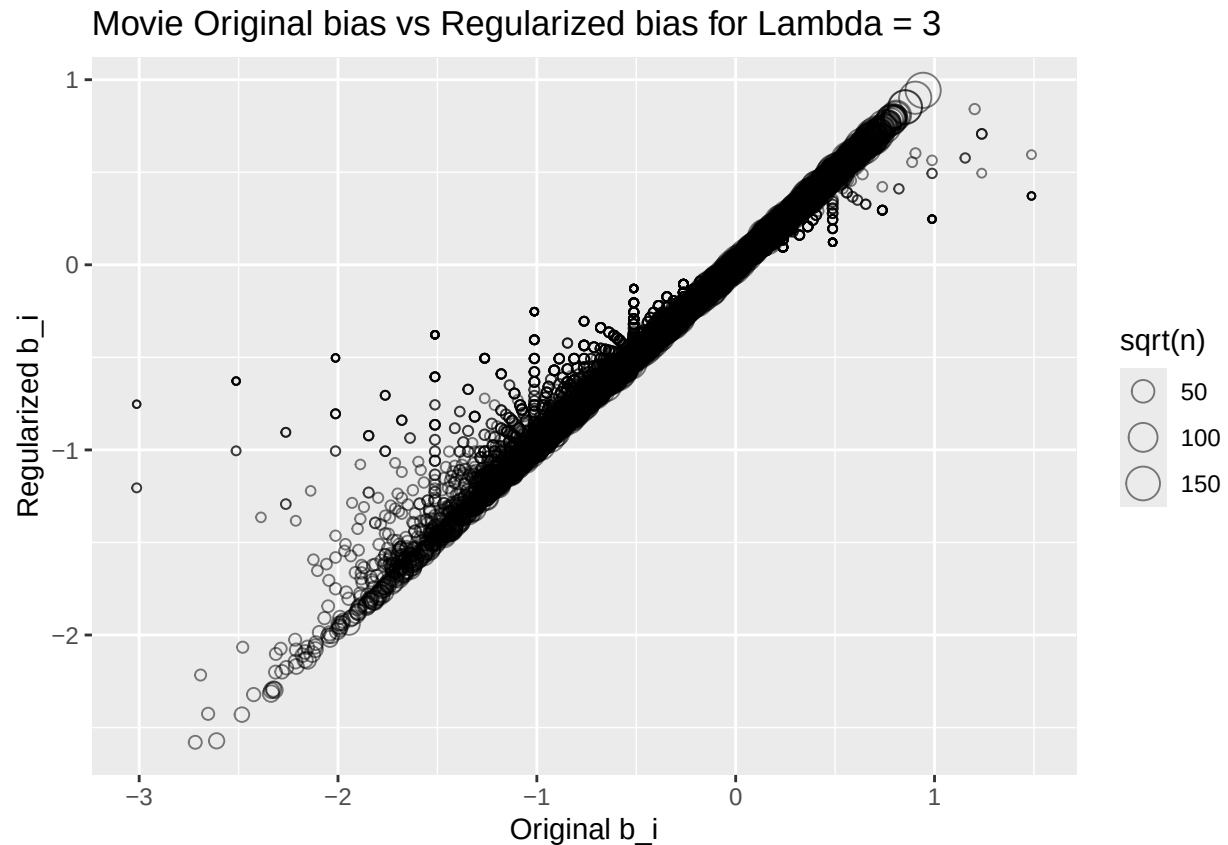
Regularization will be used to reduce the impact of large residuals that are coming from movies with very few ratings (or small sample sizes).

Using Regularization, the small sample size bias is penalized by adding a Lambda to “n”, where n is the number of ratings for movie i.

- The higher the “n”, the lower the impact of Lambda, thus, less penalized.
- The lower the “n”, the higher the impact of Lambda, thus, more penalized.

Lambda = 3 will be used to generate Movie_Regularized_bias (Regularized_b_ias_i), and compare it to Movie_bias (original_b_i) in a plot.

The following plot shows how Regularized_bias and Original_bias differ more for small values of n.



The lists of the top 15 best and worst movies are generated again but using `Movie_regularized_bias` instead of `Movie_bias`.

##

Following is the list of the 15 best movies based on `Movie_regularized_bias`.

Movie title (Best)

Shawshank Redemption, The (1994)
 Godfather, The (1972)
 Usual Suspects, The (1995)
 Schindler's List (1993)
 More (1998)
 Casablanca (1942)
 Rear Window (1954)
 Sunset Blvd. (a.k.a. Sunset Boulevard) (1950)
 Third Man, The (1949)
 Double Indemnity (1944)
 Paths of Glory (1957)
 Seven Samurai (Shichinin no samurai) (1954)
 Godfather: Part II, The (1974)
 Dark Knight, The (2008)
 Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb (1964)

##

Following is the number of ratings of the 15 best movies based just on `Movie_bias`.

```
## [1] 28015 17747 21648 23193      7 11232  7935  2922  2967  2154  1571  5190
## [13] 11920  2353 10627

##
## These movies have mostly thousands of ratings, which provides better predictors.
##
## Following is the list of the 15 worst movies based on Movie_regularized_bias.
```

Movie title (Worst)
SuperBabies: Baby Geniuses 2 (2004)
From Justin to Kelly (2003)
Pokémon Heroes (2003)
Disaster Movie (2008)
Carnosaur 3: Primal Species (1996)
Glitter (2001)
Pokemon 4 Ever (a.k.a. Pokémon 4: The Movie) (2002)
Gigli (2003)
Barney's Great Adventure (1998)
Hip Hop Witch, Da (2000)
Faces of Death: Fact or Fiction? (1999)
Yu-Gi-Oh! (2004)
Faces of Death 6 (1996)
Son of the Mask (2005)
Carnosaur 2 (1995)

```
##
## Following is the number of ratings of the 15 worst movies based just on Movie_bias.
## [1]  56 199 137  32  68 339 202 313 208  14  58  80  79 165  92
```

```
##
## These movies have mostly hundreds of ratings, which provides better predictors.
```

The prediction is $Y = \mu_{\text{hat}} + \text{Movie_Regularized_bi}$ where $\text{Movie_Regularized_bi} = \text{Movie regularized bias with } \Lambda = 3$.

RMSE can be calculated using the train set (edx) and $\mu_{\text{hat}} + \text{Movie_Regularized_bi}$ as predictor.

```
## model_regularized_bi_rmse:  0.942409
```

The fourth model and its RMSE are as follows:

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.0603313	
2: Predictor = $\mu_{\text{hat}} + \text{Movie_bi}$	0.9423475	
3: Predictor = $\mu_{\text{hat}} + \text{Movie_bi} + \text{User_bu}$	0.8567039	
4: Predictor = $\mu_{\text{hat}} + \text{Movie_regularized_bi}$, arbitrary $\Lambda = 3$	0.9424090	3

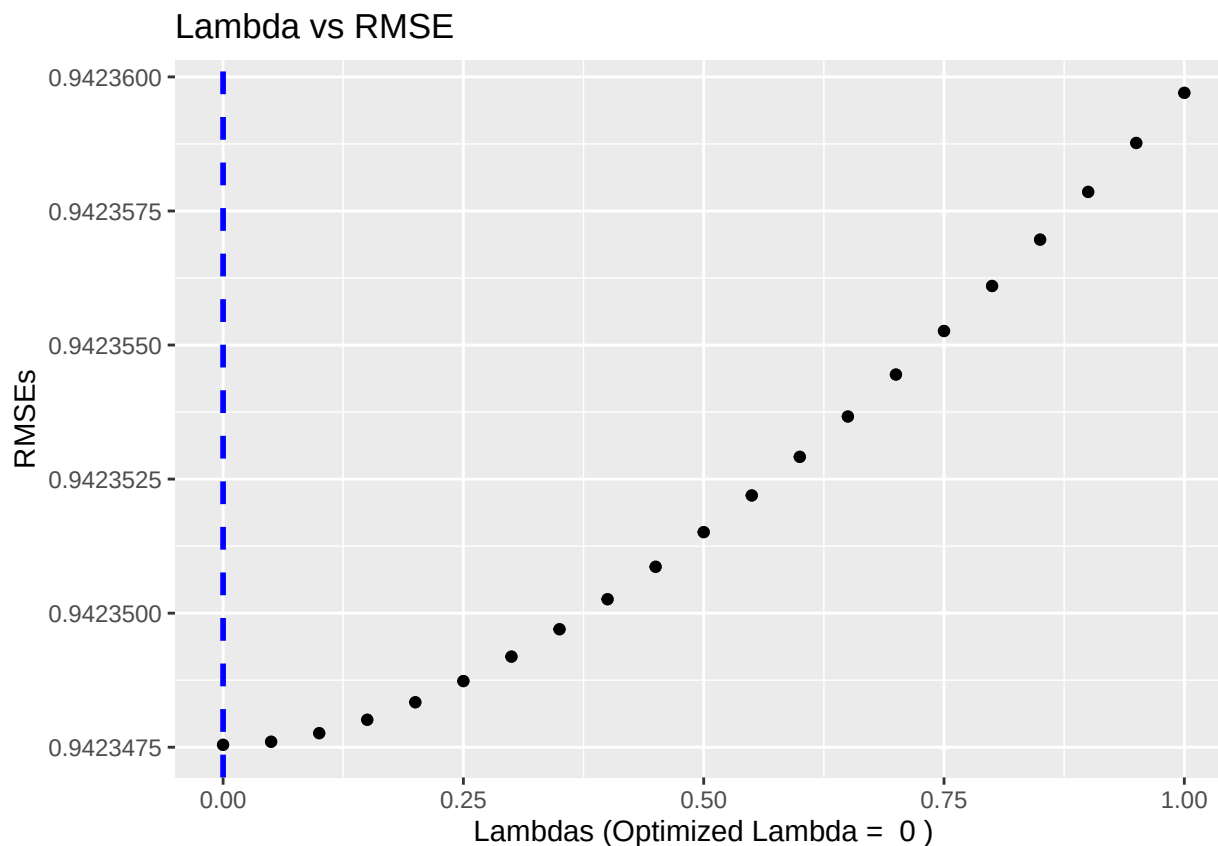
When using Movie Regularized bias instead of Movie bias, the RMSE is lower than the second model's RMSE, which has just Movie bias.

2.3.6 Fifth model: $Y = \mu_{\text{hat}} + \text{Movie_Regularized_bias}$ (Regularization) with Optimized Lamda

The fifth model will optimize the value of Lambda for the fourth model.

Values of Lambda will be used in an iteration to find the Lambda that minimizes RMSE. Lambda will be tested between 0 and 1, with an increment of 0.05.

The following plot shows Lambda versus RMSE.



The prediction is $Y = \mu_{\text{hat}} + \text{Movie_Regularized_bi}$ with Optimized Lambda where $\text{Movie_Regularized_bi} = \text{Movie regularized bias}$ with Lambda optimized between 0 and 1 at 0.05 as interval

RMSE can be calculated using the train set (edx) and $\mu_{\text{hat}} + \text{Movie_Regularized_bi}$ as predictor.

Lambda that minimizes RMSE: 0

Minimum RMSE: 0.9423475

The fifth model and its RMSE are as follows:

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.0603313	
2: Predictor = $\mu_{\text{hat}} + \text{Movie_bi}$	0.9423475	
3: Predictor = $\mu_{\text{hat}} + \text{Movie_bi} + \text{User_bu}$	0.8567039	
4: Predictor = $\mu_{\text{hat}} + \text{Movie_regularized_bi}$, arbitrary Lambda = 3	0.9424090	3
5: Predictor = $\mu_{\text{hat}} + \text{Movie_regularized_bi}$, optimized Lambda = 0	0.9423475	0

RMSE is optimized for Lambda = 0

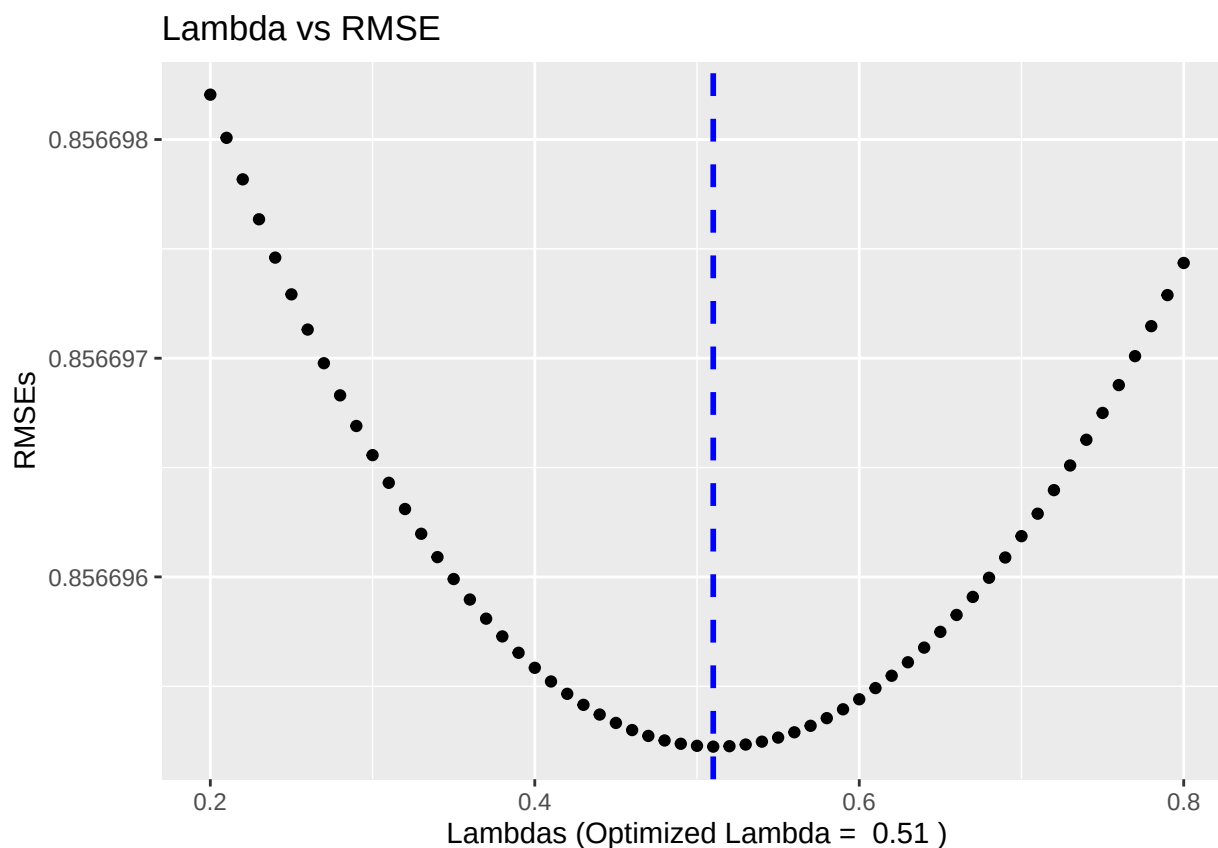
2.3.7 Sixth model: $Y = \mu_{\text{hat}} + \text{Movie_Regularized_bias} + \text{User_Regularized_bias}$ (Regularization) with Optimized Lamda

In the sixth model, User regularized bias will be added to Movie regularized bias (the previous model.) Lambda will also be optimized for a minimum RMSE.

Lambda will be tested between 0.2 and 0.8, with an increment of 0.01.

The following plot shows Lambda versus RMSE.

The following plot shows Lambda versus RMSE.



The prediction is $Y = \mu_{\text{hat}} + \text{Movie_Regularized_bi} + \text{User_Regularized_bu}$ with Optimized Lambda where Movie-Regularized_bi = Movie regularized bias; User-Regularized_bu = User Regularized bias with Lambda optimized between 0.2 and 0.8, with an increment of 0.01.

RMSE can be calculated using the train set (edx) and $\mu_{\text{hat}} + \text{Movie_Regularized_bi} + \text{User_Regularized_bu}$ as predictor.

Optimum Lambda: 0.51

Minimum RMSE: 0.8566952

The sixth model and its RMSE are as follows:

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.0603313	
2: Predictor = $\mu_{\text{hat}} + \text{Movie_bi}$	0.9423475	
3: Predictor = $\mu_{\text{hat}} + \text{Movie_bi} + \text{User_bu}$	0.8567039	
4: Predictor = $\mu_{\text{hat}} + \text{Movie_regularized_bi}$, arbitrary Lambda = 3	0.9424090	3

method	RMSE	Lambda
5: Predictor = μ_{hat} + Movie_regularized_bi, optimized Lambda = 0	0.9423475	0
6: Predictor = μ_{hat} + Movie_regularized_bi + User_regularized_bu, optimized Lambda = 0.51	0.8566952	0.51

3 RESULTS

The following table shows all six models and their RMSEs:

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.0603313	
2: Predictor = μ_{hat} + Movie_bi	0.9423475	
3: Predictor = μ_{hat} + Movie_bi + User_bu	0.8567039	
4: Predictor = μ_{hat} + Movie_regularized_bi, arbitrary Lambda = 3	0.9424090	3
5: Predictor = μ_{hat} + Movie_regularized_bi, optimized Lambda = 0	0.9423475	0
6: Predictor = μ_{hat} + Movie_regularized_bi + User_regularized_bu, optimized Lambda = 0.51	0.8566952	0.51

RMSE has been decreasing as every model incorporated additional features.

The last model incorporated regularization with both Movie bias and User bias and Lambda was optimized resulting in the lowest RMSE.

Predictor = μ_{hat} + Movie_regularized_bi + User_regularized_bu

Minimum RSME (0.8566952) was found with the Regularization model including

Movie and User biases for optimized lambda = 0.51

The sixth model has an RMSE that is lower than: 0.86490 and is the model with the lowest RMSE. This model will provide the best movie suggestions to Netflix users.

Now, the model is applied to the Test set and its RMSE is calculated.

Test set's RSME (0.8652204)

Movie and User biases for optimized lambda = 0.51

method	RMSE	Lambda
1: Predictor = μ_{hat}	1.0603313	
2: Predictor = μ_{hat} + Movie_bi	0.9423475	
3: Predictor = μ_{hat} + Movie_bi + User_bu	0.8567039	
4: Predictor = μ_{hat} + Movie_regularized_bi, arbitrary Lambda = 3	0.9424090	3
5: Predictor = μ_{hat} + Movie_regularized_bi, optimized Lambda = 0	0.9423475	0
6: Predictor = μ_{hat} + Movie_regularized_bi + User_regularized_bu, optimized Lambda = 0.51	0.8566952	0.51
7: TEST Predictor = μ_{hat} + Movie_regularized_bi + User_regularized_bu, optimized Lambda = 0.51	0.8652204	0.51

Movie_Regularized_bias and User_Regularized_bias RMSE with optimized Lambda = 0.51 applied on the test dataset has an RMSE of 0.8652204, which is slightly above the training set's RMSE and the target RMSE (0.86490) It is not uncommon that the test set RMSE has a small variance with the training set RMSE

4 CONCLUSION

In this analysis, six models were considered. Each model was based on a subset of 10 million rows of the MovieLens dataset.

The models were reducing RMSE to get to the minimum RMSE found when Regularization was applied to both Movie bias and User bias together with optimizing Lambda.

Predictor = $\mu_{\text{hat}} + \text{Movie_regularized_bi} + \text{User_regularized_bu}$

Using regularization, RMSE was minimum when Lambda was optimized through iteration. That was the Sixth model.

With this model we built a recommendation system that will predict which movies users would like to watch.

Additional analysis can be done with residuals.

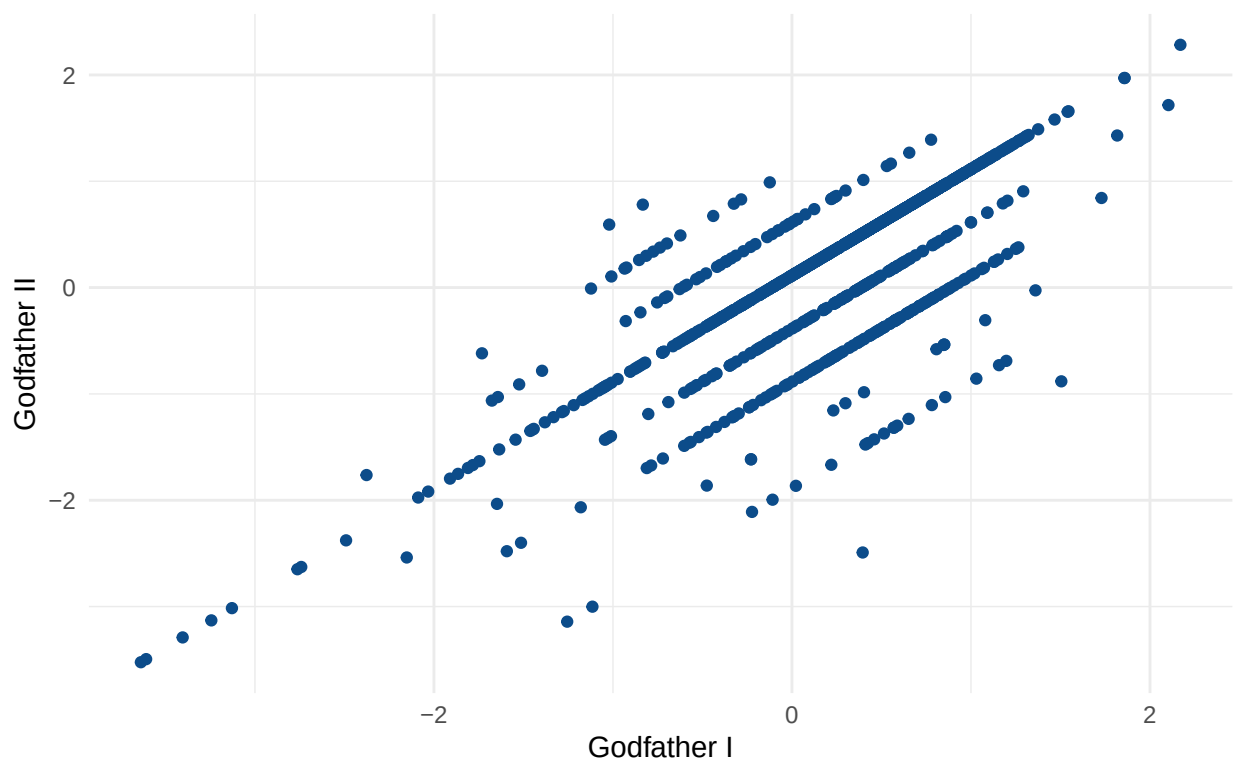
Residuals are equal to $(Y - b_i - b_u)$

Four movies were selected to analyze their residuals from $Y = \mu_{\text{hat}} + b_i + b_u$

- Godfather I
- Godfather II
- Godfather III
- Sleepless in Seattle

The following plot shows residuals for Godfather I versus residuals for Godfather II.

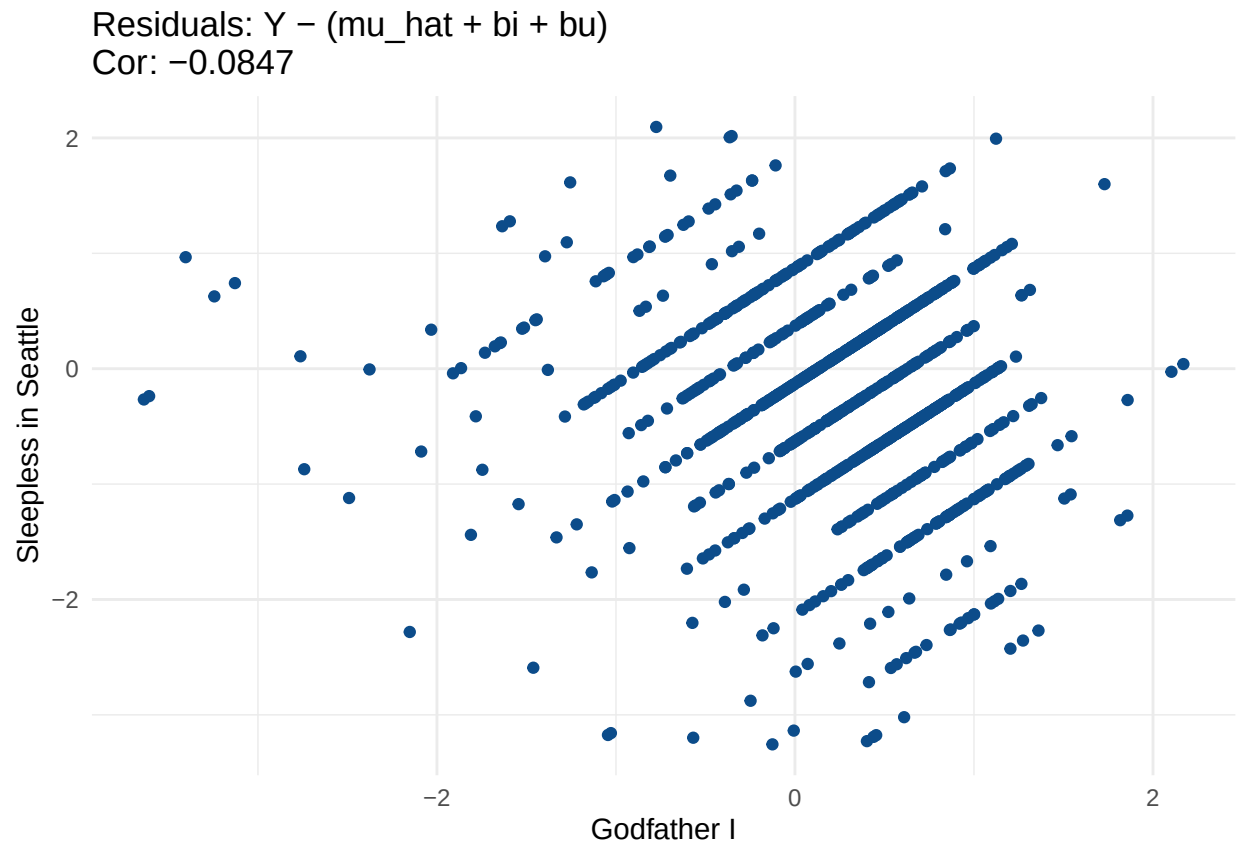
Residuals: $Y - (\mu_{\text{hat}} + b_i + b_u)$
Cor: 0.7694



There is a relationship between those two sets of residuals. The correlation shown is 0.7693758

This relationship should be incorporated in a model.

The following plot shows residuals for Godfather I versus residuals for Sleepless in Seattle.



There does not seem to be a relationship between those two sets of residuals. The correlation shown is -0.0847177, almost 0

Following is the correlation table for the residuals of the four movies.

	GF1	GF2	GF3	SIS
GF1	1.0000000	0.7693758	0.2695905	-0.0847177
GF2	0.7693758	1.0000000	0.2947621	-0.0855634
GF3	0.2695905	0.2947621	1.0000000	0.0005715
SIS	-0.0847177	-0.0855634	0.0005715	1.0000000

The correlation table shows that there is relationship between some movies' residuals:

- Godfather I and Godfather II have a correlation of 0.77
- Godfather I and Sleepless in Seattle have a correlation of -0.085, no correlation
- Godfather II and Sleepless in Seattle have a correlation of -0.085, no correlation
- Godfather III does not correlate well with any movie on the table.

It means that only some movies are related and their residuals correlated and should be incorporated in a model.

In particular Matrix factorization should further be used to build a seventh model to predict $Y = \text{Rating}$

Additional elements for further analysis are:

- a. One limitation of the analysis is that it only considered 10 million records, for future studies, it is recommended to:

- apply a larger dataset
- apply updated versions of the dataset as they are coming

b. One element that was not included in this analysis is “genre.”

“genre” can be incorporated to create a seventh model and calculate the RMSE:

Seventh model: $Y = \text{movie_bi} + \text{User_bu} + \text{Genre_bg}$